

Minimax Mean Squared Error for Low-Order Network Interference under Bernoulli Assignment

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Abstract

This paper studies design-based estimation of the finite-population all-treated-versus-all-control effect under known bounded-degree interference and low-order polynomial potential-outcome schedules. Under common-probability Bernoulli assignment, coefficient mass bounded by B , graph in- and out-degree bounded by d , and fixed interaction order β , the minimax mean squared error is characterized up to constants depending only on (β, p) :

$$B^2 \min \left\{ 1, \frac{d(k_\star(d, \beta, p))^d}{n} \right\}.$$

Here $k_\star(d, \beta, p)$ is the maximum $r \in \{1, \dots, \min\{\beta, d\}\}$ whose Bernoulli all-treated-versus-all-control contrast is nonzero when this set is nonempty, with $k_\star = 0$ under the degree-zero convention. For each fixed (β, p) , the implied constants are uniform in n , d , and B . The upper bound is achieved by the clipped SNIPE estimator, and the lower bound is obtained from complete-block least-favourable priors built from the block representer. The same block energy A_d governs a uniformly bounded-outcome model class, yielding the equivalent frontier $B^2 \min\{1, dA_d/n\}$. On complete-block graph sequences, the local linear benchmark has exact finite-sample value $B^2 A_d/m$, and asymptotic optimality is characterized by a blockwise quadratic excess criterion. For fair-coin assignment, only odd interaction orders contribute to A_d ; for first-order interactions ($\beta = 1$), the frontier becomes $B^2 \min\{1, d^2/n\}$.

1 Introduction

Network experiments routinely target contrasts between global treatment policies while observing outcomes under randomized assignments that expose only a small portion of the potential-outcome schedule. When one unit's outcome may depend on nearby units' assignments, the variance of a graph-aware estimator is shaped both by the size of each exposure neighborhood and by the number of other outcomes that share the same treatment coordinates. This paper gives a finite-population, design-based minimax calculation for that tradeoff under known bounded-degree interference and low-order polynomial response schedules.

The estimand is the average all-treated-versus-all-control contrast. The population, graph, and potential outcomes are fixed, and randomness comes from independent Bernoulli assignment with common treatment probability $p \in (0, 1)$. Each unit's outcome is a polynomial in the assignments in its graph neighborhood, with interaction order bounded by a fixed β and

*Formal verification details are summarized in the verification appendix.

coefficient mass bounded by B . The graph has maximum in-degree and out-degree at most d , with self-loops counted. This is the coefficient-mass model class in [definition 5](#), and risk is the design mean squared error in [definition 8](#).

The main result, [theorem 1](#), establishes the minimax frontier

$$R_n^*(d, \beta, p, B) \asymp_{\beta, p} B^2 \min \left\{ 1, \frac{d \binom{d}{k_*(d, \beta, p)}}{n} \right\}.$$

The index $k_*(d, \beta, p)$, defined in [definition 1](#), is the maximum $r \in \{1, \dots, \min\{\beta, d\}\}$ whose Bernoulli all-treated-versus-all-control contrast is nonzero when this set is nonempty, with $k_* = 0$ under the degree-zero convention. Thus the rate separates three components: the sample-size factor $1/n$, the graph-overlap factor d , and the number of exposed interaction subsets inside a degree- d neighborhood. The clipped SNIPE estimator in [definition 7](#) attains the upper bound, while the unprojected SNIPE estimator has worst-case mean squared error bounded by the corresponding untruncated variance scale.

The proof identifies two finite-design objects behind the rate. The first is the complete-block score energy A_d introduced in [definition 9](#). [Lemma 1](#) proves that, for complete-block product Bernoulli designs on $\{0, 1\}^d$ and the low-order polynomial space $\mathcal{P}_d$, the block score represents the all-treated-versus-all-control contrast, A_d is the minimum energy of an unbiased local weight, and A_d^{-1} is the minimum energy of a normalized local perturbation. The second is the overlap count in [theorem 1](#), which shows that each r -way treatment subset in one neighborhood can be reused by at most d outcome scores. Together these facts convert local Bernoulli information into the global variance scale dA_d/n , and [lemma 1](#) relates A_d to $\binom{d}{k_*(d, \beta, p)}$.

The lower bound uses complete directed blocks as least-favourable carriers. Within each block, the perturbation is aligned with the normalized Riesz representer h_d , and random baseline coefficients are drawn from a smooth compactly supported density. The two signed priors shift the target by $\rho\delta$ and $-\rho\delta$, while their squared Hellinger distance is controlled by $m\delta^2/(B^2A_d)$. This construction yields the lower frontier at the same scale as the SNIPE upper bound.

The complete-block calculation also gives an exact benchmark for local linear estimators. In [theorem 2](#), for a fixed complete-block graph sequence with block size d_t , $m_t = n_t/d_t$ blocks, and the moment restrictions in [definition 18](#), the local linear minimax risk is

$$R_t^{\text{loc, lin}} = \frac{B^2 A_{d_t}}{m_t}.$$

The theorem gives a finite-sample blockwise quadratic criterion $\Psi_{b,t}$, proves that its excess over $d_t^2 A_{d_t}$ exactly characterizes asymptotic attainment of the benchmark, and shows that risk-optimal local linear weights can remain separated from the canonical complete-block SNIPE score in normalized L^2 distance when $d_t \rightarrow \infty$.

The coefficient-mass model has a bounded-outcome analogue in [definition 21](#). For this class, [theorem 3](#) characterizes the minimax order through the same energy scale:

$$R_{n, \infty}^*(d, \beta, p, B) \asymp_{\beta, p} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\}.$$

The projected estimator is clipped to the natural total-effect range $[-2B, 2B]$, and the unprojected SNIPE estimator remains unbiased with worst-case mean squared error at most $B^2 dA_d/n$. On complete blocks with $d \mid n$, canonical SNIPE has exact worst-case risk $B^2 A_d/m$ for both the coefficient-mass and bounded-outcome classes.

The fair-coin specialization in [theorem 4](#) makes the exposure calculation explicit. At $p = 1/2$, even-order Bernoulli contrasts cancel and odd orders determine the energy:

$$A_d = 4 \sum_{\substack{1 \leq r \leq \bar{\beta}_d \\ r \text{ odd}}} \binom{d}{r}.$$

For first-order interactions ($\beta = 1$), $A_d = 4d$, so the minimax frontier is of order $B^2 \min\{1, d^2/n\}$. This case gives a transparent calibration of the quadratic degree cost induced by first-order interference under fair Bernoulli assignment.

The results build on the design-based potential-outcomes tradition for randomized experiments [[Rubin, 1978](#), [Horvitz and Thompson, 1952](#), [Imbens and Rubin, 2015](#)] and on interference frameworks based on exposure mappings, local dependence, and network randomization [[Sobel, 2006](#), [Hudgens and Halloran, 2008](#), [Aronow and Samii, 2017](#), [Ugander et al., 2013](#), [Eckles et al., 2017](#), [Leung, 2022](#), [Sävje et al., 2021](#), [Hu et al., 2022](#), [Gao and Ding, 2025](#)]. The closest antecedent is the SNIPE analysis for low-order neighborhood interference in [Cortez-Rodriguez et al. \[2023\]](#), which supplies the known-graph Bernoulli estimator and its overlap-aware design variance calculation. The present paper characterizes the matched minimax mean-squared-error frontier in that setting, with explicit degree dependence $dA_d/n \asymp_{\beta,p} d\binom{d}{k_\star}/n$. Its lower bound is generated by a complete-block least-favourable family aligned with the Riesz representer, and its extensions give the bounded-outcome frontier and the exact complete-block local linear benchmark. The calculation also supplies a unit-randomized reference point for recent work on low-order, clustered, and covariate-adjusted interference estimators [[Harshaw et al., 2022](#), [Eichhorn et al., 2024](#), [Wang and Li, 2026](#)]. A Lean 4 development accompanies the paper; the verification appendix states the precise formal scope.

2 Setup and Assumptions

We work in a finite-population design-based setting. The population is the finite unit set V_n with size n ; indices such as i and j refer to units in V_n . A known directed graph G_n records interference, with N_i denoting the assignment coordinates that can enter unit i 's potential outcome. For an assignment vector z , the potential outcome schedule is $Y_i(z)$, and the realized assignment is Z . The observed outcome is $Y_i^{\text{obs}} = Y_i(Z)$. The estimand is the average all-treated-versus-all-control contrast τ_n . This finite-population design-based randomization world and the accompanying fixed-potential-outcome polynomial interference world provide the objects used in all risk statements below.

The assignment mechanism is the independent Bernoulli design. We write p for the common treatment probability, P_Z for the product assignment law, and $\mathbb{E}_Z$ for expectation under that law.

The notation is grouped by role in the results.

Object group	Role	First main use
$(V_n, G_n, N_i, Z, Y_i^{\text{obs}}, \tau_n), c, (d, \beta, B, p)$	finite-population decision problem	theorem 1
$g_{i,\beta,p}, \hat{\tau}_n^{\text{SNIPE}}, \hat{\tau}_n^{\text{up}}, R_n^*$	score, estimator, and coefficient-class risk	theorem 1
$g_d, A_d, h_d, L^2(P_Z), \mathcal{P}_d, L_d$	complete-block quantities	Riesz theorems 1 and 2
$V_b, U_b, \delta, \rho, \Pi_+, \Pi_-$	least-favourable construction	block theorem 1
$G_t, n_t, d_t, m_t, b(i), w_{i,t}, \mathcal{E}_t^{\text{loc,lin}}, \Psi_{b,t}$	complete-block local linear benchmark	theorem 2
$\mathcal{M}_{n,d,\beta}^\infty(B), R_{n,\infty}^*, \hat{\tau}_{n,\infty}^{\text{up}}$	uniformly bounded-outcome comparison	theorem 3

The setup first introduces the design, graph, model class, estimator, and risk. The block programs, least-favourable laws, local-linear sequence, and bounded-outcome envelope then appear at the points where the main theorems use them.

Assumption 1. `[ass:bernoulli-design]` (Bernoulli assignment) *The assignment law P_Z is the common-probability product Bernoulli design with $p \in (0, 1)$. Equivalently, the assignments $\{Z_j : j \in V_n\}$ are mutually independent and*

$$\Pr(Z_j = 1) = p$$

for every $j \in V_n$.

[Assumption 1](#) is the standard unit-level Bernoulli randomization condition used in design-based analyses of SNIPE-type estimators [[Cortez-Rodriguez et al., 2023](#)]. Its independence and common-probability clauses make the finite-design moments of centered assignment monomials explicit.

Interference is local in the known graph. The parameter d bounds both the number of assignment coordinates entering each unit's outcome and the number of outcomes affected by each assignment coordinate, with self-loops counted in both directions.

Assumption 2. `[ass:bounded-degree]` (Bounded interference degree) *Self-loops are allowed and count in both degrees. The interference neighborhoods satisfy*

$$\max_{i \in V_n} |N_i| \leq d \quad \text{and} \quad \max_{j \in V_n} |\{i \in V_n : j \in N_i\}| \leq d.$$

[Assumption 2](#) is the standard bounded in- and out-degree neighborhood interference condition with self-loops [[Cortez-Rodriguez et al., 2023](#)]. The first bound limits the dimension of each unit's exposure polynomial, while the second controls how many summands in an estimator can share a treatment coordinate.

Potential outcomes are represented by raw assignment monomials on the relevant neighborhood. The coefficient schedule is c , with $c_{i,S}$ denoting the coefficient on a finite subset S of assignment coordinates. The interaction order β fixes the maximum monomial degree used by the model.

Assumption 3. `[ass:low-order]` (Low-order interactions) *For every $i \in V_n$ and every finite subset $S \subseteq V_n$,*

$$c_{i,S} = 0$$

whenever $|S| > \beta$.

[Assumption 3](#) is the standard bounded polynomial interaction order condition [[Cortez-Rodriguez et al., 2023](#)]. It turns the neighborhood outcome schedule into a low-order polynomial and separates the role of graph degree from the role of interaction order.

The coefficient envelope B normalizes the size of each unit's outcome polynomial.

Assumption 4. `[ass:bounded-coefficient-mass]` (Bounded coefficient mass) *For every $i \in V_n$,*

$$\sum_{S \subseteq N_i} |c_{i,S}| \leq B.$$

[Assumption 4](#) is the standard SNIPE $Y_{\max}$ coefficient-mass normalization [[Cortez-Rodriguez et al., 2023](#)]. It supplies a uniform scale for the total-effect target and for the design mean squared error comparisons.

The exposed interaction order records how much of the polynomial class can be seen within a degree- d neighborhood. The Bernoulli contrast coefficient is $\Delta_r(p)$, with r indexing interaction order; it is the order- r all-treated-versus-all-control contrast of a centered assignment monomial,

$$\Delta_r(p) := (1 - p)^r - (-p)^r.$$

Whether $\Delta_r(p)$ vanishes decides which interaction orders the design can expose, and hence the value of $k_\star$ below.

Definition 1. `[def:exposed-order]` (Exposed order $\bar{\beta}_d$ and $k_\star(d, \beta, p)$) *For every $d \geq 0$, set*

$$\bar{\beta}_d := \min\{\beta, d\}.$$

In particular, $\bar{\beta}_0 = 0$.

If $\bar{\beta}_d \geq 1$, define

$$k_\star(d, \beta, p) := \max\{r \in \{1, \dots, \bar{\beta}_d\} : \Delta_r(p) \neq 0\}.$$

Since $\Delta_1(p) = 1$, the index set contains $r = 1$, so this maximum is well defined.

If $\bar{\beta}_d = 0$, equivalently $d = 0$ or $\beta = 0$, set

$$k_\star(d, \beta, p) := 0$$

as a totality convention rather than as an attained interaction order. Thus the maximum case applies exactly when $d \geq 1$ and $\beta \geq 1$. Throughout the subsequent results, $\beta \geq 1$ is fixed.

The quantities $\bar{\beta}_d$ and $k_\star(d, \beta, p)$ determine which centered monomial contrasts carry identifying content under the Bernoulli design.

Definition 2. `[def:zero-degree-conventions]` (Zero-degree conventions $\binom{0}{0}$, g_0 , A_0 , and h_0) *For every $\beta \in \mathbb{N}$ and $p \in \mathbb{R}$, the degree-zero conventions are*

$$\bar{\beta}_0 := \min\{\beta, 0\} = 0, \quad k_\star(0, \beta, p) := 0, \quad g_0 := 0, \quad A_0 := 0.$$

The normalized representer is guarded at degree zero by setting

$$h_0 := 0, \quad h_{0,T} := 0$$

for every block-coordinate subset T of the zero-coordinate block, with h_0 imposed as an empty-object convention rather than as the quotient g_0/A_0 .

These conventions allow the formulas for scores, energies, and representers to remain total across degree values, while the main results use the positive-degree regime stated in their conditions.

We next collect the graph and coefficient classes. The graph carrier is $\mathcal{G}_{n,d}$, the coefficient carrier is $\mathcal{C}_{n,d,\beta}(B)$, and their product model class is $\mathcal{M}_{n,d,\beta}(B)$.

Definition 3. [def:graph-class] (Graph class $\mathcal{G}_{n,d}$) *The graph class with degree bound d is*

$$\mathcal{G}_{n,d} := \{G_n \subseteq V_n \times V_n : G_n \text{ satisfies } \text{assumption 2}\}.$$

Definition 4. [def:coefficient-class] (Coefficient class $\mathcal{C}_{n,d,\beta}(B)$) *The coefficient class with interaction order β and mass radius B is*

$$\mathcal{C}_{n,d,\beta}(B) := \{c = (c_{i,S})_{i \in V_n, S \subseteq V_n} : c_{i,S} = 0 \text{ whenever } S \not\subseteq N_i, \text{ } c \text{ satisfies } \text{assumptions 3 and 4}\}.$$

Definition 5. [def:model-class] (Model class $\mathcal{M}_{n,d,\beta}(B)$) *Under assumptions 2 to 4, with coefficient schedules supported on the in-neighborhoods in the sense that $c_{i,S} = 0$ whenever $S \not\subseteq N_i$, the model class is*

$$\mathcal{M}_{n,d,\beta}(B) := \{(G_n, c) : G_n \in \mathcal{G}_{n,d}, c \in \mathcal{C}_{n,d,\beta}(B)\}.$$

Together, definitions 3 to 5 specify the coefficient-mass model over which the first minimax risk is evaluated.

The estimator uses centered Bernoulli contrast scores. For unit i , the score $g_{i,\beta,p}$ weights observed outcomes by all centered neighborhood monomials up to the exposed order.

Definition 6. [def:snipe-score] (SNIPE score $g_{i,\beta,p}$) *For unit i , the SNIPE score is*

$$g_{i,\beta,p}(z) := \sum_{r=1}^{\bar{\beta}_d} \frac{\Delta_r(p)}{[p(1-p)]^r} \sum_{\substack{S \subseteq N_i \\ |S|=r}} \prod_{j \in S} (z_j - p).$$

The unprojected estimator $\hat{\tau}_n^{\text{SNIPE}}$ averages observed outcomes weighted by these scores, and the projected estimator $\hat{\tau}_n^{\text{up}}$ clips that average to $[-B, B]$ for the coefficient-mass risk comparison.

Definition 7. [def:snipe-estimator] (SNIPE estimator $\hat{\tau}_n^{\text{SNIPE}}$) *The unprojected SNIPE estimator is*

$$\hat{\tau}_n^{\text{SNIPE}} := \frac{1}{n} \sum_{i \in V_n} Y_i^{\text{obs}} g_{i,\beta,p}(Z).$$

Its projected version is

$$\hat{\tau}_n^{\text{up}} := \Pi_{[-B,B]}(\hat{\tau}_n^{\text{SNIPE}}),$$

where $\Pi_{[-B,B]}$ denotes Euclidean projection onto $[-B, B]$.

The decision problem is graph-aware and design-based: estimators may use the graph, the assignment, and the observed outcomes, while risk averages only over the random assignment. The generic estimator is $\hat{\tau}_n$, and the resulting coefficient-class minimax risk is $R_n^*(d, \beta, p, B)$. This is the design-based decision-theoretic world used by the main theorem.

Definition 8. `[def:minimax-risk]` (Minimax risk $R_n^*(d, \beta, p, B)$) The minimax design mean squared error over $\mathcal{M}_{n,d,\beta}(B)$ is

$$R_n^*(d, \beta, p, B) := \inf_{\hat{\tau}_n} \sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z \left[(\hat{\tau}_n - \tau_n)^2 \right],$$

where the infimum ranges over estimators $\hat{\tau}_n$ measurable with respect to $(G_n, Z, \{Y_i^{\text{obs}} : i \in V_n\})$.

The lower-bound construction and the local linear benchmark both reduce to complete blocks. For one d -block, the block score g_d , energy A_d , normalized representer h_d , block-coordinate subset T , and raw coefficient $h_{d,T}$ are as follows.

Definition 9. `[def:block-score-energy]` (Complete-block score and energy g_d, A_d, h_d) For $d \geq 1$ and a Bernoulli assignment vector $z = (z_1, \dots, z_d)$, define

$$g_d(z) := \sum_{r=1}^{\bar{\beta}_d} \frac{\Delta_r(p)}{[p(1-p)]^r} \sum_{\substack{S \subseteq \{1, \dots, d\} \\ |S|=r}} \prod_{j \in S} (z_j - p).$$

The corresponding energy is

$$A_d := \sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} \frac{\Delta_r(p)^2}{[p(1-p)]^r}.$$

The normalized block representer is

$$h_d := \frac{g_d}{A_d}.$$

Its raw-monomial expansion is

$$h_d(z) = \sum_{T \subseteq \{1, \dots, d\}} h_{d,T} \prod_{j \in T} z_j.$$

The least-favourable family places complete directed blocks inside the population. Its notation includes the number of blocks m , active units q , active fraction ρ , active block V_b , block index b , baseline halfwidth s , density f_s , block baseline U_b , representer mass bound $H_{\beta,p}$, perturbation amplitude δ , sign σ , and prior Π_σ . This is the least-favourable block world used to calibrate the minimax lower bound.

Definition 10. `[def:block-family]` (Least-favourable block family Π_σ) For $B > 0$ and $1 \leq d \leq n$, set

$$m := \lfloor n/d \rfloor, \quad q := md, \quad \rho := q/n.$$

Partition q units into m blocks V_b of size d . Each block contains every within-block directed arrow, including loops, and the remaining $n - q$ units are isolated.

Set

$$s := B/2, \quad f_s(u) := s^{-1} \cos^2\left(\frac{\pi u}{2s}\right) \mathbf{1}\{|u| \leq s\}.$$

Let $U_1, \dots, U_m$ be independent draws from density f_s . Define

$$H_{\beta,p} := \sup_{d \geq 1} \sum_T |h_{d,T}|$$

and choose

$$\delta := B \min\{(2H_{\beta,p})^{-1}, (4\pi)^{-1}\} \min\{1, \sqrt{A_d/m}\}.$$

Under Π_σ , each active unit $i \in V_b$ has coefficients

$$c_{i,T} := U_b \mathbf{1}\{T = \emptyset\} + \sigma \delta h_{d,T}, \quad |T| \leq k_\star.$$

All remaining coefficients, including coefficients of inactive units, are zero. The variables U_b randomize fixed potential-outcome schedules as prior draws over schedules.

The two signed versions of the construction also induce prior-predictive laws. The negative law is named separately because the affinity comparison in the lower-bound argument uses the pair symmetrically.

Definition 11. [synth_4] (Negative least-favourable prior-predictive law Π_-) We write Π_- for the prior-predictive law induced by the least-favourable block construction with sign $\sigma = -1$: draw $Z \sim \prod_{j \in V_n} \text{Bernoulli}(p)$ and independent $U_1, \dots, U_m$ with density f_s , set the active coefficients in block V_b to $c_{i,T} = U_b \mathbf{1}\{T = \emptyset\} - \delta h_{d,T}$ for $|T| \leq k_\star$ and all remaining coefficients to zero, and take the induced joint law of $(Z, \{Y_i^{\text{obs}} : i \in V_n\})$.

Definition 12. [synth_3] (Positive score-aligned block prior Π_+) In the complete-block construction with $B > 0$ and $1 \leq d \leq n$, we write $\Pi_+(n, d, \beta, B, p)$ for the probability law on fixed coefficient schedules induced by independent draws $U_b \sim f_s$ under which each active unit $i \in V_b$ has $c_{i,T} = U_b \mathbf{1}\{T = \emptyset\} + \delta h_{d,T}$ for $|T| \leq k_\star(d, \beta, p)$, and every other coefficient, including each coefficient of an inactive unit, equals 0.

For the complete-block Riesz calculations, the finite design space and the polynomial contrast space are useful in their own right. We use the block-local Riesz world: the block polynomial subspace $\mathcal{P}_d$, candidate polynomial f , candidate perturbation h , all-one-versus-all-zero contrast functional L_d , candidate block-local weight w , and the associated minimum-energy programs. The contrast functional evaluates a block polynomial at the all-treated and all-control assignments,

$$L_d(f) := f(\mathbf{1}_d) - f(\mathbf{0}_d),$$

so the normalization $L_d(h) = 1$ below fixes the scale at which a perturbation moves the target. Writing A_i for the local SNIPE energy of unit i , the same construction applied to that unit's own neighborhood gives

$$A_i := \sum_{r=1}^{\min\{\beta, |N_i|\}} \binom{|N_i|}{r} \frac{\Delta_r(p)^2}{[p(1-p)]^r},$$

which reduces to the complete-block energy A_d when $|N_i| = d$.

Definition 13. [synth_2] (Design $L^2(P_Z)$ space) We write $L^2(P_Z)$ for the real vector space of functions $w : \{0, 1\}^{\{1, \dots, d\}} \rightarrow \mathbb{R}$ with finite design second moment $\mathbb{E}_Z[w(Z)^2] < \infty$ under the product Bernoulli assignment law P_Z with success probability p .

Definition 14. [synth_1] (Block polynomial space $\mathcal{P}_d$) For $d \geq 1$, $\mathcal{P}_d$ is the linear subspace of $L^2(P_Z)$ spanned by the raw monomials $z \mapsto \prod_{j \in S} z_j$ with $S \subseteq \{1, \dots, d\}$ and $|S| \leq \bar{\beta}_d$.

Definition 15. `[def:perturbation-program]` (Perturbation program $\text{PerturbProg}_{d,\beta,p}$) For $d \geq 1$, define

$$\text{PerturbProg}_{d,\beta,p} := \min_{h \in \mathcal{P}_d} \mathbb{E}_Z[h(Z)^2]$$

subject to

$$L_d(h) = 1.$$

Definition 16. `[def:weight-program]` (Weight program $\text{WeightProg}_{d,\beta,p}$) For $d \geq 1$, define

$$\text{WeightProg}_{d,\beta,p} := \min_{w \in L^2(P_Z)} \mathbb{E}_Z[w(Z)^2]$$

subject to

$$\mathbb{E}_Z[w(Z)] = 0$$

and, for every nonempty $S \subseteq \{1, \dots, d\}$ with $|S| \leq \bar{\beta}_d$,

$$\mathbb{E}_Z \left[w(Z) \prod_{j \in S} Z_j \right] = 1.$$

The perturbation program $\text{PerturbProg}_{d,\beta,p}$ and the weight program $\text{WeightProg}_{d,\beta,p}$ express the same complete-block information through dual normalizations: one fixes the all-one-versus-all-zero contrast and minimizes polynomial energy, while the other fixes unbiased raw-monomial moments and minimizes weight energy.

The fixed-graph local linear benchmark specializes the population sequence to complete blocks. Let G_t be the fixed complete-block graph at index t , with carrier V_{n_t} , size n_t , block size d_t , and neighborhoods $N_i^{G_t}$. The local coefficient class is $\mathcal{M}_t(G_t)$.

Definition 17. `[def:local-linear-class]` (Fixed-graph class $\mathcal{M}_t(G_t)$) Under [assumptions 3](#) and [4](#), on the fixed graph G_t , define $\mathcal{M}_t(G_t)$ as the set of coefficient schedules c satisfying

$$c_{i,S} = 0$$

whenever $S \subseteq N_i^{G_t}$ and $|S| \leq \bar{\beta}_{d_t}$ fail to hold, and satisfying, for every $i \in V_{n_t}$,

$$\sum_{S \subseteq N_i^{G_t}} |c_{i,S}| \leq B.$$

A local linear estimator uses a block-local assignment weight $w_{i,t}$. If $b(i)$ denotes the block containing unit i , then $Z_{b(i)}$ is the assignment vector in that block, $\hat{\tau}_{w,t}$ is the resulting estimator, and $\mathcal{E}_t^{\text{loc},\text{lin}}$ is the moment-restricted class.

Definition 18. `[def:local-linear-estimator-class]` (Local linear estimators $\mathcal{E}_t^{\text{loc},\text{lin}}$) Define $\mathcal{E}_t^{\text{loc},\text{lin}}$ as the set of estimators

$$\hat{\tau}_{w,t} := n_t^{-1} \sum_{i \in V_{n_t}} w_{i,t}(Z_{b(i)}) Y_i(Z)$$

such that $w_{i,t} \in L^2(P_Z)$ is measurable with respect to $Z_{b(i)}$,

$$\mathbb{E}_Z[w_{i,t}] = 0,$$

and, for every nonempty $S \subseteq N_i^{G_t}$ with $|S| \leq \bar{\beta}_{d_t}$,

$$\mathbb{E}_Z \left[w_{i,t} \prod_{j \in S} Z_j \right] = 1.$$

The fixed-graph local linear risk $R_t^{\text{loc,lin}}$ evaluates these estimators over the corresponding coefficient schedules.

Definition 19. [def:local-linear-risk] (Local linear risk $R_t^{\text{loc,lin}}$) For a fixed complete-block graph G_t , define

$$R_t^{\text{loc,lin}} := \inf_{\hat{\tau}_{w,t} \in \mathcal{E}_t^{\text{loc,lin}}} \sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z \left[(\hat{\tau}_{w,t} - \tau_{n_t})^2 \right].$$

The supremum ranges over coefficient schedules on the fixed graph G_t .

Finally, we introduce the uniformly bounded-outcome class. Its schedule carrier is $\mathcal{C}_{n,d,\beta}^\infty(B)$, and its model class is $\mathcal{M}_{n,d,\beta}^\infty(B)$.

Definition 20. [def:bounded-outcome-coefficient-class] (Bounded-outcome coefficient class $\mathcal{C}_{n,d,\beta}^\infty(B)$) Under [assumption 3](#), define

$$\mathcal{C}_{n,d,\beta}^\infty(B) := \left\{ c : c_{i,S} = 0 \text{ whenever } S \not\subseteq N_i, \quad c_{i,S} = 0 \text{ whenever } |S| > \beta, \quad \sup_{z \in \{0,1\}^{V_n}} |Y_i(z)| \leq B \text{ for every } i \in V_n \right\}$$

Definition 21. [def:bounded-outcome-model-class] (Bounded-outcome model class $\mathcal{M}_{n,d,\beta}^\infty(B)$) Under [assumptions 2 and 3](#), define

$$\mathcal{M}_{n,d,\beta}^\infty(B) := \{ (G_n, c) : G_n \in \mathcal{G}_{n,d}, \quad c \in \mathcal{C}_{n,d,\beta}^\infty(B) \}.$$

The notation for the two minimax risks distinguishes the coefficient-mass class from the uniformly bounded-outcome class. We write $R_{n,\ell_1}^*(d, \beta, p, B)$ for the coefficient-mass risk and $R_{n,\infty}^*(d, \beta, p, B)$ for the bounded-outcome risk.

Definition 22. [def:two-class-minimax-risks] (Two minimax risks R_{n,ℓ_1}^* and $R_{n,\infty}^*$) For $0 \leq p \leq 1$, define

$$R_{n,\ell_1}^*(d, \beta, p, B) := R_n^*(d, \beta, p, B)$$

and

$$R_{n,\infty}^*(d, \beta, p, B) := \inf_{\hat{\tau}_n} \sup_{(G_n, c) \in \mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z \left[(\hat{\tau}_n - \tau_n)^2 \right].$$

The infimum ranges over graph-aware estimators $\hat{\tau}_n$ of G_n , Z , and $\{Y_i^{\text{obs}} : i \in V_n\}$ that, for each fixed G_n and Z , are measurable as functions of the observed-outcome vector and whose modelwise bounded-outcome risk range

$$\left\{ \mathbb{E}_Z \left[(\hat{\tau}_n - \tau_n)^2 \right] : (G_n, c) \in \mathcal{M}_{n,d,\beta}^\infty(B) \right\}$$

is bounded above.

For the bounded-outcome comparison, the SNIPE estimator is projected to the natural range of the total-effect contrast. The clipped version is $\hat{\tau}_{n,\infty}^{\text{up}}$.

Definition 23. [def:bounded-outcome-clipped-snipe] (Bounded-outcome clipped SNIPE $\hat{\tau}_{n,\infty}^{\text{up}}$) Define

$$\hat{\tau}_{n,\infty}^{\text{up}} := \Pi_{[-2B, 2B]}(\hat{\tau}_n^{\text{SNIPE}}),$$

where $\Pi_{[-2B, 2B]}$ denotes Euclidean projection onto $[-2B, 2B]$.

Two prior laws are compared through the squared Hellinger distance, used throughout in its unhalved form: for laws P and Q with densities f and g with respect to a common dominating measure λ ,

$$H^2(P, Q) := \int \left(\sqrt{f} - \sqrt{g} \right)^2 d\lambda.$$

Every constant reported for a least-favourable pair below refers to this normalization; the halved convention would rescale those constants by a factor of two.

The definitions above fix the design, model classes, estimators, block programs, and risk criteria used in the results that follow.

3 Main Results

The definitions in [definitions 5, 7 and 8](#) lead to a design-based risk comparison in which the graph is known, treatment is assigned independently, and performance is evaluated uniformly over the coefficient-mass class. The first result characterizes the degree dependence of that comparison. Its rate is governed by the exposed Bernoulli contrast order $k_*(d, \beta, p)$, equivalently by the leading nonzero centered monomial order exposed by the assignment design when a positive-order exposed contrast is present and by the degree-zero convention otherwise. The projected SNIPE estimator from [definition 7](#) attains the same order as the lower bound, while the unprojected estimator has the corresponding nontruncated variance bound.

Theorem 1. [thm:degree-frontier] (Degree Frontier Bounds) Fix $\beta \geq 1$ and $p \in (0, 1)$. Under the notation in [definitions 1, 5 and 8](#), there exist constants $c_{\beta, p}$ and $C_{\beta, p}$ such that $0 < c_{\beta, p} \leq C_{\beta, p}$ and, for every $n, d \in \mathbb{N}$ and $B \in \mathbb{R}$ with $n \geq 1$, $d \leq n$, and $B \geq 0$, the following statements hold:

- (Minimax frontier.)

$$c_{\beta, p} B^2 \min \left\{ 1, \frac{d(k_*(d, \beta, p))}{n} \right\} \leq R_n^*(d, \beta, p, B).$$

Moreover,

$$R_n^*(d, \beta, p, B) \leq \sup_{(G, c) \in \mathcal{M}_{n, d, \beta}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{up}} - \tau_n)^2 \right] \leq C_{\beta, p} B^2 \min \left\{ 1, \frac{d(k_*(d, \beta, p))}{n} \right\}.$$

The unprojected SNIPE estimator also satisfies

$$\sup_{(G, c) \in \mathcal{M}_{n, d, \beta}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] \leq C_{\beta, p} B^2 \frac{d(k_*(d, \beta, p))}{n}.$$

- (Score energy.) For every $(G, c) \in \mathcal{M}_{n, d, \beta}(B)$ and every unit $i \in V_n$,

$$A_i \leq A_d.$$

- **(Overlap counts.)** For every $(G, c) \in \mathcal{M}_{n,d,\beta}(B)$, every unit $i \in V_n$, and every integer $r \geq 1$,

$$\sum_{\ell \in V_n} \binom{|N_i \cap N_\ell|}{r} = \sum_{\substack{S \subseteq N_i \\ |S|=r}} \#\{\ell \in V_n : S \subseteq N_\ell\} \leq d \binom{|N_i|}{r} \leq d \binom{d}{r}.$$

- **(Block construction.)** If $B > 0$ and $d \geq 1$, let $m = \lfloor n/d \rfloor$, let ρ be the active population fraction, and let δ be the least-favourable perturbation amplitude. For every block-coefficient vector $U = (U_1, \dots, U_m)$ satisfying $|U_b| \leq B/2$ for all b , there exist two models M_+ and M_- in $\mathcal{M}_{n,d,\beta}(B)$, both supported on the complete d -block graph, whose coefficient schedules are the corresponding $+1$ and -1 block schedules, and whose total treatment effects satisfy

$$\tau_n(M_+) = \rho\delta, \quad \tau_n(M_-) = -\rho\delta.$$

The associated block prior densities satisfy

$$H^2(\Pi_+, \Pi_-) \leq \frac{4\pi^2 m \delta^2}{B^2 A_d}.$$

In addition,

$$\frac{1}{2} \leq \rho \leq 1, \quad \frac{A_d}{m} = \frac{dA_d/n}{\rho}.$$

[Theorem 1](#) identifies the relevant effective dimension as a degree-weighted count of exposed interactions. The upper bound combines unbiasedness of the SNIPE score with the overlap count in the theorem, so the risk grows with the number of units whose scores can share treatment coordinates. The lower bound is calibrated on complete blocks: the perturbation is aligned with the normalized block representer of [definition 9](#), and the Hellinger bound makes the target separation comparable to the same block energy. Thus the minimax comparison measures the exact price of estimating the total treatment effect under known bounded-degree interference and low-order polynomial response schedules.

The next result isolates what local linear estimators can achieve on a fixed complete-block sequence. This benchmark is useful because complete blocks are the least-favourable carriers used above and because local linear estimators are a natural graph-local analogue of inverse-probability weighted estimators in randomized experiments. The characterization is exact at each t , and the asymptotic statements distinguish attainment of the optimal block risk from closeness to the canonical SNIPE score itself.

Theorem 2. [\[thm:sharp-local-linear-constant-and-representers\]](#) (Sharp Local Linear Representers) Fix $\beta \in \mathbb{N}$, $p \in (0, 1)$, and $B > 0$, with $\beta \geq 1$. Suppose that:

- **(Block sizes.)** For each t , n_t and d_t are positive integers and $d_t \mid n_t$. Write $m_t := n_t/d_t$.
- **(Block graph.)** For each t , G_t is the directed complete-block graph on V_{n_t} : units are partitioned into m_t active blocks V_b of size d_t , each block contains all directed edges including self-loops, and there are no edges between distinct blocks.
- **(Assignment law.)** For each t , the finite design D_t satisfies [assumption 1](#) with common treatment probability p .

Then, for every t ,

$$R_t^{\text{loc,lin}} = \frac{B^2 A_{d_t}}{m_t} = \frac{B^2 d_t A_{d_t}}{n_t}.$$

For every t and every $w \in \mathcal{E}_t^{\text{loc,lin}}$,

$$\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z [(\hat{\tau}_{w,t} - \tau_{n_t})^2] = \frac{B^2}{n_t^2} \sum_{b=1}^{m_t} \Psi_{b,t}(w),$$

where, with $Z_S := \prod_{j \in S} Z_j$,

$$\Psi_{b,t}(w) := \max_{\substack{(\varepsilon_i, S_i)_{i \in V_b} \\ \varepsilon_i \in \{-1, 1\}, S_i \subseteq V_b, |S_i| \leq \bar{\beta}_{d_t}}} \mathbb{E}_Z \left[\left(\sum_{i \in V_b} \varepsilon_i \{w_{i,t}(Z_b) Z_{S_i} - \mathbf{1}(S_i \neq \emptyset)\} \right)^2 \right].$$

Moreover, for every t , every $w \in \mathcal{E}_t^{\text{loc,lin}}$, and every block V_b ,

$$d_t^2 A_{d_t} \leq \Psi_{b,t}(w).$$

For every sequence $w_t \in \mathcal{E}_t^{\text{loc,lin}}$,

$$\frac{\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z [(\hat{\tau}_{w_t,t} - \tau_{n_t})^2]}{B^2 A_{d_t}/m_t} \longrightarrow 1$$

if and only if

$$\frac{1}{m_t d_t^2 A_{d_t}} \sum_{b=1}^{m_t} \{ \Psi_{b,t}(w_t) - d_t^2 A_{d_t} \} \longrightarrow 0.$$

For every sequence $w_t \in \mathcal{E}_t^{\text{loc,lin}}$, the condition

$$\frac{1}{n_t A_{d_t}} \sum_{i \in V_{n_t}} \mathbb{E}_Z \left[\{w_{i,t}(Z_{b(i)}) - g_{d_t}(Z_{b(i)})\}^2 \right] \longrightarrow 0$$

implies

$$\frac{\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z [(\hat{\tau}_{w_t,t} - \tau_{n_t})^2]}{B^2 A_{d_t}/m_t} \longrightarrow 1.$$

If $d_t \rightarrow \infty$, then there exists a sequence $w_t \in \mathcal{E}_t^{\text{loc,lin}}$ such that

$$\frac{\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z [(\hat{\tau}_{w_t,t} - \tau_{n_t})^2]}{B^2 A_{d_t}/m_t} \longrightarrow 1,$$

and, for all sufficiently large t ,

$$\frac{1}{n_t A_{d_t}} \sum_{i \in V_{n_t}} \mathbb{E}_Z \left[\{w_{i,t}(Z_{b(i)}) - g_{d_t}(Z_{b(i)})\}^2 \right] = 2.$$

For every sufficiently large t , the same sequence also satisfies, for every bijection $\pi : V_{n_t} \rightarrow V_{n_t}$ that preserves block membership,

$$\frac{1}{n_t A_{d_t}} \sum_{i \in V_{n_t}} \mathbb{E}_Z \left[\{w_{i,t}(Z_{\pi(b(i))}) - g_{d_t}(Z_{\pi(b(i))})\}^2 \right] = 2.$$

Theorem 2 gives the exact complete-block constant $B^2 A_{d_t}/m_t$. The block criterion $\Psi_{b,t}$ is the finite-dimensional worst-case quadratic form induced by the moment restrictions in [definition 18](#); averaging its excess over $d_t^2 A_{d_t}$ is necessary and sufficient for asymptotic minimaxity within the local linear class. The final clauses show that score closeness is a sufficient route to the benchmark, while the benchmark also admits asymptotically optimal local linear weights whose normalized distance from the canonical complete-block score is fixed at two along growing block sizes. The result therefore separates the risk-optimality property from uniqueness of the SNIPE representer as a weight function.

The coefficient-mass condition in [definition 5](#) has a direct bounded-outcome counterpart in [definition 21](#). The next theorem compares the two decision problems and shows that the same complete-block energy controls the minimax order under the uniformly bounded-outcome envelope. This formulation aligns the SNIPE variance calculation with the design-based bounded-outcome risk scale used in network-experiment analyses such as [Cortez-Rodriguez et al. \[2023\]](#) and related low-order interference work [\[Harshaw et al., 2022, Wang and Li, 2026\]](#).

Theorem 3. [\[thm:bounded-outcome-degree-frontier\]](#) (Bounded-Outcome Degree Frontier) *Fix an interaction order $\beta \geq 1$ and a treatment probability $p \in (0, 1)$. There exist real constants $c_{\beta,p}$ and $C_{\beta,p}$ with*

$$0 < c_{\beta,p} \leq C_{\beta,p}$$

such that, for every $n \geq 1$, every design D satisfying [assumption 1](#) with common treatment probability p , every integer $d \leq n$, and every $B \geq 0$, the following statements hold.

- **(Model comparison.)** *The coefficient-mass class $\mathcal{M}_{n,d,\beta}(B)$ of [definition 5](#) is contained, carrier by carrier, in the uniformly bounded-outcome class $\mathcal{M}_{n,d,\beta}^\infty(B)$ of [definition 21](#). If $B > 0$ and $d \geq 1$, this containment is strict.*
- **(Frontier bounds.)** *With A_d denoting the complete-block score energy,*

$$c_{\beta,p} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\} \leq R_{n,\ell_1}^*(d, \beta, p, B),$$

$$R_{n,\ell_1}^*(d, \beta, p, B) = R_n^*(d, \beta, p, B),$$

$$R_{n,\ell_1}^*(d, \beta, p, B) \leq R_{n,\infty}^*(d, \beta, p, B),$$

and

$$R_{n,\infty}^*(d, \beta, p, B) \leq \sup_{\mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z \left[(\hat{\tau}_{n,\infty}^{\text{up}} - \tau_n)^2 \right] \leq C_{\beta,p} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\}.$$

The coefficient-class clipped SNIPE estimator satisfies

$$\sup_{\mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z \left[(\hat{\tau}_n^{\text{up}} - \tau_n)^2 \right] \leq C_{\beta,p} B^2 \min \left\{ 1, \frac{dA_d}{n} \right\}.$$

- **(Unclipped SNIPE.)** *On both $\mathcal{M}_{n,d,\beta}(B)$ and $\mathcal{M}_{n,d,\beta}^\infty(B)$, the SNIPE estimator $\hat{\tau}_n^{\text{SNIPE}}$ is unbiased under D , and its worst-case mean squared error is at most*

$$B^2 \frac{dA_d}{n}.$$

- **(Local energy and binomial scale.)** For every unit i , the local SNIPE energy on either model class is bounded by A_d . Moreover, with $k_\star = k_\star(d, \beta, p)$ as in [definition 1](#),

$$c_{\beta,p} d \binom{d}{k_\star} \leq dA_d \leq C_{\beta,p} d \binom{d}{k_\star}.$$

- **(Exact block risk.)** If $d \geq 1$ and $d \mid n$, let $m = n/d$ and let the graph be the disjoint union of m complete directed d -blocks, including loops. Then canonical SNIPE has exact worst-case risk

$$\frac{B^2 A_d}{m}$$

on the fixed block graph for both model classes and globally over both model classes. Equivalently,

$$\frac{B^2 A_d}{m} = B^2 \frac{dA_d}{n}.$$

- **(Least-favourable block family.)** If $B > 0$ and $d \geq 1$, set $m = \lfloor n/d \rfloor$, $\rho = md/n$, and let δ be the perturbation amplitude. For every vector $U = (U_1, \dots, U_m)$ with $|U_b| \leq B/2$ for all b , both signed block schedules belong to $\mathcal{M}_{n,d,\beta}(B)$ on the complete-block graph, and their targets are

$$\rho\delta \quad \text{and} \quad -\rho\delta.$$

The squared Hellinger distance between the two corresponding block priors is bounded by

$$\frac{4\pi^2 m \delta^2}{B^2 A_d},$$

and

$$\frac{1}{2} \leq \rho \leq 1, \quad \frac{A_d}{m} = \frac{dA_d/n}{\rho}.$$

[Theorem 3](#) expresses the bounded-outcome comparison through the energy A_d , and [theorem 1](#) then converts that energy to the binomial scale indexed by $k_\star(d, \beta, p)$. The bounded-outcome lower frontier is inherited from the coefficient-mass lower frontier because the coefficient-mass class is contained carrier by carrier in the bounded-outcome class, giving $R_{n,\ell_1}^\star \leq R_{n,\infty}^\star$. The strict containment statement records how the two envelopes relate on the same graph carriers. The clipped estimator $\hat{\tau}_{n,\infty}^{\text{up}}$ attains the bounded-outcome rate after projection to the natural total-effect range, while the unprojected estimator remains unbiased and has worst-case mean squared error $B^2 d A_d / n$. On complete blocks with $d \mid n$, the theorem also gives the exact canonical SNIPE risk $B^2 A_d / m$, matching the local linear value in [theorem 2](#).

The final result specializes the energy calculation to fair-coin assignment. When $p = 1/2$, even-order Bernoulli contrasts vanish and odd orders determine the complete-block score energy. For first-order interactions ($\beta = 1$), this gives the transparent rate $B^2 \min\{1, d^2/n\}$, a useful benchmark for comparing low-order network interference estimators with the overlap-aware constructions in [Cortez-Rodriguez et al. \[2023\]](#) and subsequent degree-sensitive analyses [[Harshaw et al., 2022](#), [Eichhorn et al., 2024](#), [Wang and Li, 2026](#)].

Theorem 4. `[thm:fair-coin-energy-frontier]` (Fair-Coin Energy Frontier) *Let $p = 1/2$. The following statements hold.*

- **(Fair-coin contrasts and exposed order.)** For every $n, d, \beta \in \mathbb{N}$ and $B \in \mathbb{R}$ such that $n \geq 1$, $1 \leq d \leq n$, $\beta \geq 1$, and $B \geq 0$, the contrast coefficients and energy in [definition 1](#) satisfy, for every $r \geq 1$,

$$\Delta_r(1/2) = \begin{cases} 2^{1-r}, & r \text{ odd}, \\ 0, & r \text{ even}, \end{cases}$$

and

$$A_d = 4 \sum_{\substack{1 \leq r \leq \bar{\beta}_d \\ r \text{ odd}}} \binom{d}{r}.$$

Moreover, $k_*(d, \beta, 1/2)$ is the largest odd integer no greater than $\bar{\beta}_d$, with value 0 when no such odd integer exists.

- **(Degree-one minimax frontier.)** There exist constants $c_{\text{lower}}, c_{\text{upper}} \in \mathbb{R}$ with

$$0 < c_{\text{lower}} \leq c_{\text{upper}}$$

such that, for every $n, d \in \mathbb{N}$ and $B \in \mathbb{R}$ satisfying $n \geq 1$, $1 \leq d \leq n$, and $B \geq 0$, the degree-one energy is

$$A_d = 4d,$$

and the minimax risks in [definition 8](#) obey

$$c_{\text{lower}} B^2 \min \left\{ 1, \frac{d^2}{n} \right\} \leq R_n^*(d, 1, 1/2, B) = R_{n, \ell_1}^*(d, 1, 1/2, B) \leq R_{n, \infty}^*(d, 1, 1/2, B) \leq c_{\text{upper}} B^2 \min \left\{ 1, \frac{d^2}{n} \right\}.$$

If $d \mid n$, the unprojected SNIPE estimator satisfies the exact worst-case identities

$$\sup_{\mathcal{M}_{n, d, 1}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] = \frac{4B^2 d^2}{n}$$

and

$$\sup_{\mathcal{M}_{n, d, 1}^\infty(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] = \frac{4B^2 d^2}{n}.$$

[Theorem 4](#) makes the role of parity explicit. At $p = 1/2$, the exposed order is the largest odd interaction order up to $\bar{\beta}_d$, and the block energy is four times the corresponding sum of odd binomial coefficients. In the first-order interaction case ($\beta = 1$), $A_d = 4d$, so the degree factor in [theorem 3](#) becomes $dA_d = 4d^2$. The resulting minimax rate $B^2 \min\{1, d^2/n\}$ shows how first-order interference converts the graph degree into a quadratic design cost under fair-coin Bernoulli assignment.

4 Discussion and Extensions

The results above identify the design-based cost of total-effect estimation through a small set of finite-population quantities. The common feature in [theorems 1, 3](#) and [4](#) is the product of an overlap factor and a block energy. Under the bounded-degree condition, each treatment coordinate can enter at most d unit-level scores, and the overlap count in [theorem 1](#) turns this graph fact into the variance scale dA_d/n . The energy A_d then records how much Bernoulli

design information is required to recover the all-treated-versus-all-control contrast from low-order centered monomials.

This interpretation connects the rate to the exposure and overlap structure emphasized in network-experiment analyses [Leung, 2022, Sävje et al., 2021, Hu et al., 2022, Gao and Ding, 2025]. In the present model, the known graph fixes the neighborhoods and the low-order polynomial restriction fixes which assignment interactions are relevant. The exposed order $k_*(d, \beta, p)$ summarizes the maximum positive contrast order visible under Bernoulli assignment when such an order exists, with $k_* = 0$ under the degree-zero convention, and [theorem 1](#) converts that order into the binomial scale $d \binom{d}{k_*(d, \beta, p)} / n$. For fixed β and p , the rate therefore separates the sample-size contribution $1/n$, the graph-overlap contribution d , and the interaction-order contribution through the number of exposed subsets inside a degree- d neighborhood.

The closest estimator-level antecedent, Cortez-Rodriguez et al. [2023], develops SNIPE for low-order neighborhood interference under known-graph Bernoulli assignment and analyzes the resulting design variance through exposure overlap. The minimax frontier here turns that variance scale into a matched decision-theoretic characterization: the complete-block lower bound, the clipped SNIPE upper bound, and the binomial degree formula $dA_d/n \asymp_{\beta, p} d \binom{d}{k_*} / n$ all use the same block energy. The bounded-outcome result and the local-linear complete-block theorem provide the corresponding bounded-envelope and graph-local benchmarks.

The complete-block benchmark clarifies why the same quantity appears in both upper and lower bounds. In the least-favourable construction of [theorem 1](#), complete directed blocks align all within-block neighborhoods, so the block representer in [definition 9](#) generates the hardest local contrast at energy A_d . On this graph, [theorem 2](#) gives the exact local linear value $B^2 A_d / m_t$, equivalently $B^2 d_t A_d / n_t$. The global SNIPE upper bound attains the same order on arbitrary graphs because the score uses the same blockwise Riesz structure locally, while the bounded out-degree condition controls how often those local scores can share assignment coordinates.

The local linear characterization also separates risk optimality from a unique choice of weights. The canonical complete-block score is sufficient for attaining the benchmark, and [theorem 2](#) gives the exact blockwise quadratic criterion whose average excess characterizes asymptotic minimaxity inside the local linear class. Thus the complete-block calculation is more than a lower-bound device: it identifies the precise variance target that any asymptotically optimal local linear rule must match, while allowing multiple weight sequences to attain that target. This perspective complements recent work on low-order and degree-sensitive interference estimators [Wang and Li, 2026, Eichhorn et al., 2024].

The bounded-outcome extension in [theorem 3](#) shows that the same degree dependence is calibrated by A_d under a uniform potential-outcome envelope. The projection interval changes from the coefficient-mass scale to the natural total-effect range, but the unprojected SNIPE variance remains governed by $B^2 d A_d / n$, and clipping converts that variance calculation into the bounded risk comparison. The fair-coin specialization in [theorem 4](#) makes this calibration explicit: for $p = 1/2$, odd interaction orders determine A_d , and first-order interference yields the rate $B^2 \min\{1, d^2/n\}$.

Limitations and future work. The analysis is developed for known bounded-degree interference graphs, independent Bernoulli assignment with common treatment probability, and noiseless finite-population potential outcomes represented by low-order polynomial schedules. Natural extensions include alternative randomization schemes, covariate adjustment, graph uncertainty, clustered assignment, and sampling or outcome noise. These directions would connect

the minimax calculation here with broader design and exposure-mapping frameworks for network experiments [Leung, 2022, Sävje et al., 2021, Hu et al., 2022, Gao and Ding, 2025].

Appendices

A Proofs and Auxiliary Lemmas

This appendix collects the auxiliary finite-design facts that support the main risk comparisons. The first statement is the complete-block Riesz calculation used in the least-favourable construction and in the local linear benchmark. The second statement is the graph-counting identity that converts bounded in- and out-degree into the overlap scale appearing in [theorems 1](#) and [3](#). Together, these two ingredients isolate the two sources of the rate: block energy under Bernoulli assignment and neighborhood overlap in the graph.

Lemma 1. [`lem:block-energy-representer`] (Block Energy Representer) *Fix an integer $\beta \geq 1$ and $p \in (0, 1)$. With $k_\star(d, \beta, p)$ as in [definition 1](#), there exist constants $c_1, c_2, H \in \mathbb{R}$, depending on (β, p) , such that $0 < c_1 \leq c_2$ and $0 < H$, and the following statements hold for every integer $d \geq 1$:*

- **(Energy scale.)**

$$c_1 \binom{d}{k_\star(d, \beta, p)} \leq A_d \leq c_2 \binom{d}{k_\star(d, \beta, p)}.$$

- **(Representer identities.)** *For every finite design D on $\{0, 1\}^d$ satisfying the common- p product Bernoulli assignment condition in [assumption 1](#), and every $f \in \mathcal{P}_d$,*

$$\mathbb{E}_D[g_d(Z)f(Z)] = L_d(f).$$

For the same D ,

$$L_d(h_d) = 1, \quad \mathbb{E}_D[h_d(Z)^2] = A_d^{-1}.$$

- **(Perturbation program.)** *For every finite design D on $\{0, 1\}^d$ satisfying the common- p product Bernoulli assignment condition in [assumption 1](#), the representer h_d is feasible for the perturbation program in [definition 15](#). For every feasible perturbation h ,*

$$A_d^{-1} \leq \mathbb{E}_D[h(Z)^2],$$

with equality if and only if $h = h_d$.

- **(Weight program.)** *For every finite design D on $\{0, 1\}^d$ satisfying the common- p product Bernoulli assignment condition in [assumption 1](#), and every weight w feasible for the weight program in [definition 16](#) under D ,*

$$A_d \leq \mathbb{E}_D[w(Z)^2],$$

with equality if and only if $w = g_d$.

- **(Program values.)** *For every pair of witnesses $0 \leq p$ and $p \leq 1$,*

$$\text{PerturbProg}_{d, \beta, p} = A_d^{-1}, \quad \text{WeightProg}_{d, \beta, p} = A_d.$$

- (*Coefficient envelope.*)

$$\sum_{T \subseteq \{1, \dots, d\}} |h_{d,T}| \leq H.$$

Proof of lemma 1. Fix $\beta \geq 1$ and $p \in (0, 1)$. For $1 \leq r \leq \beta$, write

$$e_r(p) := \frac{\Delta_r(p)^2}{[p(1-p)]^r}, \quad m_r(p) := \frac{|\Delta_r(p)|}{[p(1-p)]^r} (1+p)^r.$$

Let c_1 be the minimum of $e_r(p)$ over the exposed orders $1 \leq r \leq \beta$ with $\Delta_r(p) \neq 0$, let $c_2 := 2^\beta \sum_{r=1}^\beta e_r(p)$, and let

$$H := c_1 + \frac{2^\beta \sum_{r=1}^\beta m_r(p)}{c_1}.$$

Since $\Delta_1(p) = 1$ and $p(1-p) > 0$, this exposed set contains 1, and hence $c_1 > 0$. Applying the uniform comparison at $d = 1$, together with $\binom{1}{k_\star(1, \beta, p)} > 0$, gives $c_1 \leq c_2$. The displayed formula for H is c_1 plus a nonnegative term, so $H > 0$.

Now fix $d \geq 1$. The energy comparison is obtained by taking $k = k_\star(d, \beta, p)$: the lower bound isolates the k -term in A_d , while the upper bound writes A_d as the finite sum over $1 \leq r \leq \bar{\beta}_d$ and uses the fixed-order domination

$$\binom{d}{r} \leq 2^\beta \binom{d}{k}$$

for every exposed $r \leq k$, with zero contrast terms contributing zero.

Let D be any common- p product Bernoulli design on $\{0, 1\}^d$. The score g_d pairs to one with every eligible nonconstant raw monomial and has mean zero on the constant monomial; by linearity over the raw-monomial span $\mathcal{P}_d$, for every $f \in \mathcal{P}_d$,

$$\mathbb{E}_D[g_d(Z)f(Z)] = L_d(f).$$

The same identities give $L_d(g_d) = A_d$ and $\mathbb{E}_D[g_d(Z)^2] = A_d$, and the positivity of A_d follows from the $r = 1$ summand. Since $h_d = g_d/A_d$, this yields

$$L_d(h_d) = 1, \quad \mathbb{E}_D[h_d(Z)^2] = A_d^{-1}.$$

The centered-monomial expansion of g_d in the raw basis places g_d , and therefore h_d , in $\mathcal{P}_d$; together with $L_d(h_d) = 1$, this proves feasibility for the perturbation program. If $h \in \mathcal{P}_d$ is feasible, then $\mathbb{E}_D[g_d(Z)h(Z)] = L_d(h) = 1$ and $\mathbb{E}_D[g_d(Z)^2] = A_d$. Expanding the nonnegative square gives

$$\mathbb{E}_D[(A_d h(Z) - g_d(Z))^2] = A_d^2 \mathbb{E}_D[h(Z)^2] - A_d,$$

so $A_d^{-1} \leq \mathbb{E}_D[h(Z)^2]$. Equality forces the displayed square to have expectation zero; the strict product Bernoulli law has positive mass on every assignment, so $A_d h = g_d$ pointwise, equivalently $h = h_d$. The converse equality follows from the preceding identity for h_d .

For the weight program, feasibility extends from the raw constraints to all of $\mathcal{P}_d$, and $g_d \in \mathcal{P}_d$. Hence every feasible w satisfies $\mathbb{E}_D[w(Z)g_d(Z)] = L_d(g_d) = A_d$, while $\mathbb{E}_D[g_d(Z)^2] = A_d$. Expanding

$$\mathbb{E}_D[(w(Z) - g_d(Z))^2] = \mathbb{E}_D[w(Z)^2] - A_d$$

gives $A_d \leq \mathbb{E}_D[w(Z)^2]$, with equality exactly when $w = g_d$ by the same positive-mass argument.

The canonical feasible perturbation h_d and canonical feasible weight g_d give the matching attainable values in the two finite-dimensional programs, and the two optimality inequalities just proved give the reverse inequalities for every admissible pair of endpoint witnesses $0 \leq p$ and $p \leq 1$. Thus

$$\text{PerturbProg}_{d,\beta,p} = A_d^{-1}, \quad \text{WeightProg}_{d,\beta,p} = A_d.$$

Finally, expand each centered product in g_d into raw monomials and apply the triangle inequality. This bounds the raw coefficient mass of h_d by

$$A_d^{-1} \sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} m_r(p).$$

The same fixed-order binomial comparison and the lower energy bound $c_1\left(\binom{d}{k_\star(d,\beta,p)}\right) \leq A_d$ reduce this quantity to at most H . $\square$

Lemma 1 supplies the finite-dimensional normalization used throughout the block arguments. The identity $\mathbb{E}_D[g_d(Z)f(Z)] = L_d(f)$ states that the complete-block score represents the all-one-versus-all-zero contrast on the low-order polynomial space from [definition 14](#). The perturbation and weight programs in [definitions 15](#) and [16](#) are therefore dual descriptions of the same Bernoulli information: the normalized perturbation has energy A_d^{-1} , while the unbiased block weight has energy A_d . The coefficient envelope gives a uniform way to embed the score-aligned perturbation inside the coefficient-mass model class in [definition 10](#). These calculations are the block analogue of inverse-probability and Horvitz–Thompson moment identities for randomized experiments [[Horvitz and Thompson, 1952](#), [Aronow and Samii, 2017](#)].

The next lemma records the overlap count used to pass from block-local variance identities to arbitrary bounded-degree graphs. It is purely combinatorial: a shared r -tuple of treatment coordinates contributes to the covariance of two unit-level scores only when that tuple lies in both neighborhoods, and the out-degree bound controls how often each tuple can be reused.

Lemma 2. [[lem:overlap-count](#)] (Overlap Count Bound) *Let G be a graph on V_n . Suppose that*

- *(Bounded degree.) G satisfies [assumption 2](#) with degree bound d .*
- *(Order.) $r \in \mathbb{N}$ satisfies $1 \leq r$.*
- *(Unit.) $i \in V_n$.*

Then

$$\sum_{l \in V_n} \binom{|N_i \cap N_l|}{r} = \sum_{\substack{S \subseteq N_i \\ |S|=r}} \#\{l \in V_n : S \subseteq N_l\} \leq d \binom{|N_i|}{r} \leq d \binom{d}{r}.$$

Proof of [lemma 2](#). Let

$$\mathcal{F} := \{S \subseteq N_i : |S| = r\}.$$

For each $l \in V_n$, the binomial coefficient $\binom{|N_i \cap N_l|}{r}$ counts the r -subsets of $N_i \cap N_l$, equivalently

$$\binom{|N_i \cap N_l|}{r} = \sum_{S \in \mathcal{F}} \mathbf{1}\{S \subseteq N_l\}.$$

Summing over l and commuting the two finite sums gives

$$\sum_{l \in V_n} \binom{|N_i \cap N_l|}{r} = \sum_{S \in \mathcal{F}} \#\{l \in V_n : S \subseteq N_l\}.$$

Fix $S \in \mathcal{F}$. Since $|S| = r$ and $1 \leq r$, choose $j \in S$. Whenever $S \subseteq N_l$, this j belongs to N_l , so the set of l 's with $S \subseteq N_l$ is contained in the out-neighborhood of j . The out-degree part of [assumption 2](#) therefore gives

$$\#\{l \in V_n : S \subseteq N_l\} \leq d.$$

Summing this bound over $S \in \mathcal{F}$ and using $|\mathcal{F}| = \binom{|N_i|}{r}$ yields

$$\sum_{S \in \mathcal{F}} \#\{l \in V_n : S \subseteq N_l\} \leq d \binom{|N_i|}{r}.$$

Finally, the in-degree part of [assumption 2](#) gives $|N_i| \leq d$, and monotonicity of binomial coefficients in the upper argument gives

$$d \binom{|N_i|}{r} \leq d \binom{d}{r}.$$

Combining the displayed identities and inequalities proves the claim. $\square$

[Lemma 2](#) is the graph-theoretic step behind the SNIPE upper bound. After expanding products of centered Bernoulli monomials, only common neighborhood coordinates contribute nonzero covariance terms under [assumption 1](#). The equality rewrites the sum by first choosing the shared subset and then counting affected units; the final inequalities use the two degree bounds in [assumption 2](#). This converts local score energy into the global variance scale dA_d/n , matching the overlap-aware variance calculations used in network-interference estimators [[Cortez-Rodriguez et al., 2023](#), [Harshaw et al., 2022](#)].

The proof of [theorem 1](#) combines the two auxiliary lemmas. For the upper bound, the SNIPE score is unbiased for each low-order unit contrast by the same centered-monomial orthogonality encoded in [lemma 1](#); the covariance expansion is then summed with [lemma 2](#). Projection to $[-B, B]$ preserves the target scale of [definition 8](#) and turns the unprojected variance bound into the truncated risk comparison. For the lower bound, the complete-block family in [definition 10](#) uses h_d as the score-aligned perturbation, so the two signed priors separate the target by $\rho\delta$ while their affinity is controlled by the energy A_d^{-1} .

The proof of [theorem 2](#) evaluates the same block program inside the restricted local linear class. The moment equations defining [definition 18](#) reduce each block to the weight program in [definition 16](#), and [lemma 1](#) gives the minimum second moment A_{d_t} . The quadratic criterion $\Psi_{b,t}$ then records the exact blockwise worst-case contribution of admissible weights, yielding the finite-sample value and the asymptotic characterization stated in [theorem 2](#).

For the bounded-outcome comparison in [theorem 3](#), the coefficient-mass class embeds carrier by carrier into the uniformly bounded-outcome class, so the lower bound from [theorem 1](#) transfers through the risk ordering in [definition 22](#). The upper bound uses the same unbiased SNIPE variance calculation, with projection to $[-2B, 2B]$ as defined in [definition 23](#). On complete blocks, [lemma 1](#) gives the exact canonical variance $B^2 A_d/m$, aligning the global bounded-outcome comparison with the local linear benchmark.

Finally, [theorem 4](#) follows by evaluating the Bernoulli contrast coefficients at $p = 1/2$. Even-order contrasts cancel, odd-order contrasts remain, and the energy formula in [definition 9](#) reduces to the stated odd-binomial sum. In the first-order interaction case ($\beta = 1$), this gives $A_d = 4d$, so the general degree-dependent comparison in [theorem 3](#) specializes to the rate $B^2 \min\{1, d^2/n\}$.

B Verification Note

This appendix records the formal verification scope for the mathematical claims stated in the paper. The theorem statements and their proofs are machine-checked in Lean 4 for the finite Bernoulli design algebra underlying [assumption 1](#), the polynomial-interference constructions in [assumption 3](#) and [definition 5](#), the SNIPE unbiasedness and variance calculations associated with [definitions 6](#) and [7](#), and the minimax-risk definitions in [definitions 8](#) and [22](#). The same checked development covers the least-favourable complete-block family in [definition 10](#), the continuous-prior integrated-risk step, the affinity-based total-variation bound used for the block lower bound, and the block Riesz programs in [definitions 15](#) and [16](#).

The verification also covers the formal statements of [theorems 1](#) to [4](#) and the auxiliary statements collected in the proof appendix. The bibliographic inputs and econometric interpretation are supplied by the cited randomized-experiment, potential-outcomes, design-based, SNIPE, and low-order interference literatures [[Horvitz and Thompson, 1952](#), [Rubin, 1978](#), [Imbens and Rubin, 2015](#), [Aronow and Samii, 2017](#), [Cortez-Rodriguez et al., 2023](#), [Harshaw et al., 2022](#)].

C Proofs of the main results

Proof of [theorem 2](#). 1. Fix t and write $n = n_t$, $d = d_t$, and $m = m_t$. On the complete-block graph, every active block is a full neighborhood of size d , and $n = md$. The canonical block-local weights

$$w_{i,t}^0(Z) = g_d(Z_{b(i)})$$

belong to $\mathcal{E}_t^{\text{loc,lin}}$. The block program identities in [lemma 1](#), applied block by block, give value A_d per unit score and therefore $d^2 A_d$ per block in the block-extremal criterion. Thus every admissible local-linear rule has worst-case risk at least

$$\frac{B^2}{n^2} \sum_{b=1}^m d^2 A_d = \frac{B^2 A_d}{m},$$

while the canonical rule attains this value. Since $n = md$,

$$\frac{B^2 A_d}{m} = \frac{B^2 d A_d}{n}.$$

This proves the finite minimax identity.

2. For a fixed $w \in \mathcal{E}_t^{\text{loc,lin}}$, the squared error decomposes across complete blocks. Maximizing the coefficient signs and low-order monomials independently on each block gives

$$\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z[(\hat{\tau}_{w,t} - \tau_{n_t})^2] = \frac{B^2}{n_t^2} \sum_{b=1}^{m_t} \Psi_{b,t}(w),$$

where the finite maximum in $\Psi_{b,t}$ is the same block-extreme functional after translating the sign coding to $\varepsilon_i \in \{-1, 1\}$ and writing block monomials as Z_{S_i} .

3. The per-block lower bound follows from the all-empty monomial candidate and the block score identity. Let g be the common complete-block score on V_b . For every $i \in V_b$, blockwise unbiasedness gives

$$\mathbb{E}_Z[w_{i,t}(Z_b)g(Z_b)] = A_{d_t}, \quad \mathbb{E}_Z[g(Z_b)^2] = A_{d_t}.$$

Hence

$$0 \leq \mathbb{E}_Z \left[\left\{ \sum_{i \in V_b} w_{i,t}(Z_b) - d_t g(Z_b) \right\}^2 \right] = \mathbb{E}_Z \left[\left\{ \sum_{i \in V_b} w_{i,t}(Z_b) \right\}^2 \right] - d_t^2 A_{d_t}.$$

The all-empty choice in $\Psi_{b,t}$ contains the first expectation on the right, so

$$d_t^2 A_{d_t} \leq \Psi_{b,t}(w).$$

4. Define

$$Y_t := \frac{1}{m_t d_t^2 A_{d_t}} \sum_{b=1}^{m_t} \{\Psi_{b,t}(w_t) - d_t^2 A_{d_t}\}.$$

The preceding finite identity gives, for every t ,

$$\frac{\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z[(\hat{\tau}_{w_t,t} - \tau_{n_t})^2]}{B^2 A_{d_t}/m_t} = 1 + Y_t.$$

Therefore convergence of the risk ratio to 1 is equivalent to convergence of Y_t to 0, by subtracting and adding the constant sequence 1.

5. Let

$$X_t := \frac{1}{n_t A_{d_t}} \sum_{i \in V_{n_t}} \mathbb{E}_Z[\{w_{i,t}(Z_{b(i)}) - g_{d_t}(Z_{b(i)})\}^2].$$

The block-excess sum is nonnegative by the lower bound above. The blockwise distance inequality gives

$$Y_t \leq 2\sqrt{X_t} + X_t,$$

after using $n_t = m_t d_t$ and simplifying

$$\frac{2\sqrt{d_t^2 A_{d_t}} \sqrt{m_t} \sqrt{d_t E_t} + d_t E_t}{m_t d_t^2 A_{d_t}} = 2\sqrt{\frac{E_t}{n_t A_{d_t}}} + \frac{E_t}{n_t A_{d_t}}.$$

Thus $X_t \rightarrow 0$ implies $Y_t \rightarrow 0$ by squeezing, and the identity in the previous step gives the asserted convergence of the risk ratio.

6. Assume $d_t \rightarrow \infty$. For all sufficiently large t , $\beta < d_t$. Define w_t by adding to the canonical weights a centered full-block perturbation on one distinguished unit in each block:

$$w_{i,t}(Z) = g_{d_t}(Z_{b(i)}) + \mathbf{1}\{i = i_b^*\} \left\{ \frac{2d_t A_{d_t}}{(p(1-p))^{d_t}} \right\}^{1/2} \prod_{j \in V_{b(i)}} (Z_j - p).$$

The full-block centered monomial has order d_t , so it is orthogonal under [assumption 1](#) to every monomial of order at most β ; consequently these weights remain in $\mathcal{E}_t^{\text{loc}, \text{lin}}$. The

exact identity above gives the lower bound 1 for the risk ratio. The witness block bound gives, eventually,

$$\Psi_{b,t}(w_t) \leq d_t^2 A_{d_t} + 2\sqrt{d_t^2 A_{d_t}} \sqrt{2d_t A_{d_t}} + 2d_t A_{d_t},$$

and therefore

$$\frac{\sup_{c \in \mathcal{M}_t(G_t)} \mathbb{E}_Z[(\hat{\tau}_{w_t,t} - \tau_{n_t})^2]}{B^2 A_{d_t}/m_t} \leq 1 + 2\sqrt{\frac{2}{d_t}} + \frac{2}{d_t}.$$

Since $d_t \rightarrow \infty$, squeezing proves the displayed risk-ratio convergence for this sequence.

7. For the same sufficiently large t , the perturbation has squared energy $2d_t A_{d_t}$ on the distinguished unit of each block and zero perturbation energy on the other units. Hence

$$\sum_{i \in V_{n_t}} \mathbb{E}_Z[\{w_{i,t}(Z_{b(i)}) - g_{d_t}(Z_{b(i)})\}^2] = m_t(2d_t A_{d_t}),$$

and division by $n_t A_{d_t} = m_t d_t A_{d_t}$ gives the value 2. If π preserves block membership, the complete-block score is invariant under the induced within-block relabeling, and the full-block centered perturbation has the same squared energy after the relabeling. The same calculation gives the relabeled normalized distance 2. $\square$

Proof of [theorem 4](#). 1. Fix admissible n, d, β, B . For every $r \geq 1$,

$$\Delta_r(1/2) = (1/2)^r - (-1/2)^r = 2^{-r}\{1 - (-1)^r\}.$$

Thus $\Delta_r(1/2) = 2^{1-r}$ when r is odd, and $\Delta_r(1/2) = 0$ when r is even.

2. Expanding A_d from [definition 9](#) at $p = 1/2$ gives

$$A_d = \sum_{r=1}^{\bar{\beta}_d} \binom{d}{r} \frac{\Delta_r(1/2)^2}{((1/2)(1-1/2))^r}.$$

The preceding calculation makes the even terms vanish. For odd r ,

$$\frac{\Delta_r(1/2)^2}{((1/2)(1-1/2))^r} = \frac{(2^{1-r})^2}{(1/4)^r} = 4.$$

Therefore

$$A_d = 4 \sum_{\substack{1 \leq r \leq \bar{\beta}_d \\ r \text{ odd}}} \binom{d}{r}.$$

The exposed set in [definition 1](#) is exactly the set of odd integers in $\{1, \dots, \bar{\beta}_d\}$. Hence $k_\star(d, \beta, 1/2)$ is its maximum, with the stated zero value when the set is empty.

3. Apply [theorem 3](#) with $\beta = 1$ and $p = 1/2$. It supplies constants a, C with $0 < a \leq C$. Set

$$c_{\text{lower}} := a, \quad c_{\text{upper}} := 4C.$$

Since $C > 0$, these constants satisfy $0 < c_{\text{lower}} \leq c_{\text{upper}}$. For fixed admissible n, d, B , use the common fair Bernoulli product design on V_n , which satisfies [assumption 1](#).

4. The first-order interaction energy follows from the fair-coin energy identity. Since $d \geq 1$, $\bar{\beta}_d = \min\{1, d\} = 1$, and the odd orders in $\{1\}$ consist only of 1. Hence

$$A_d = 4 \binom{d}{1} = 4d.$$

Put $x = d^2/n$. Then $x \geq 0$, and

$$\frac{dA_d}{n} = \frac{d(4d)}{n} = 4x.$$

Moreover $\min\{1, x\} \leq \min\{1, 4x\}$.

5. The lower bound from [theorem 3](#) gives

$$aB^2 \min\left\{1, \frac{dA_d}{n}\right\} \leq R_{n, \ell_1}^*(d, 1, 1/2, B),$$

and the same result identifies

$$R_{n, \ell_1}^*(d, 1, 1/2, B) = R_n^*(d, 1, 1/2, B).$$

Using $\min\{1, x\} \leq \min\{1, 4x\}$ and $dA_d/n = 4x$ yields

$$c_{\text{lower}} B^2 \min\left\{1, \frac{d^2}{n}\right\} \leq R_n^*(d, 1, 1/2, B).$$

The class comparison in [theorem 3](#) supplies

$$R_{n, \ell_1}^*(d, 1, 1/2, B) \leq R_{n, \infty}^*(d, 1, 1/2, B).$$

For the upper bound, the same result bounds the bounded-outcome minimax risk by the clipped bounded-outcome SNIPE worst risk and then by

$$CB^2 \min\left\{1, \frac{dA_d}{n}\right\} = CB^2 \min\{1, 4x\}.$$

If $x \leq 1$, then either $4x \leq 1$, in which case $\min\{1, 4x\} = 4x$, or $4x > 1$, in which case $\min\{1, 4x\} = 1 \leq 4x$. If $x > 1$, then $\min\{1, 4x\} \leq 1 \leq 4 \min\{1, x\}$. Thus

$$\min\{1, 4x\} \leq 4 \min\{1, x\},$$

and consequently

$$R_{n, \infty}^*(d, 1, 1/2, B) \leq c_{\text{upper}} B^2 \min\left\{1, \frac{d^2}{n}\right\}.$$

Combining the displayed identities and inequalities gives the full minimax chain.

6. Finally assume $d \mid n$. The exact complete-block identities from [theorem 3](#) give, for the unprojected SNIPE estimator,

$$\sup_{\mathcal{M}_{n, d, 1}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] = \frac{B^2 A_d}{n/d} = \frac{B^2 d A_d}{n},$$

and the same identity over $\mathcal{M}_{n, d, 1}^\infty(B)$. Substituting $A_d = 4d$ gives both asserted values,

$$\frac{B^2 d A_d}{n} = \frac{4B^2 d^2}{n}.$$

□

Proof of [theorem 1](#). 1. Apply [theorem 3](#) with the fixed β and p . It gives constants a, b with $0 < a \leq b$ and, for every admissible n, d, B , the block-energy lower bound, the projected-SNIPE upper bound, the unprojected-SNIPE upper bound, the local-energy comparison, the exposed-order comparison

$$a d \binom{d}{k_*(d, \beta, p)} \leq dA_d \leq b d \binom{d}{k_*(d, \beta, p)},$$

and the block-mixture conclusions. Set

$$c_{\beta, p} := \min\{1, a^2\}, \quad C_{\beta, p} := \max\{1, b^2\}.$$

Then $0 < c_{\beta, p} \leq C_{\beta, p}$.

2. Fix $n, d \in \mathbb{N}$ and $B \in \mathbb{R}$ with $n \geq 1$, $d \leq n$, and $B \geq 0$, and write

$$x := \frac{d \binom{d}{k_*(d, \beta, p)}}{n}, \quad y := \frac{dA_d}{n}.$$

The common- p product Bernoulli design satisfies [assumption 1](#). Since $n \geq 1$, the exposed-order comparison from the preceding item gives

$$ax \leq y \leq bx,$$

and hence

$$\min\{1, ax\} \leq \min\{1, y\} \leq \min\{1, bx\}.$$

Also $\min\{1, a^2\} \leq a$: if $a \leq 1$, then $\min\{1, a^2\} \leq a^2 \leq a$, while if $a > 1$, then $\min\{1, a^2\} \leq 1 \leq a$. Splitting on $x \leq 1$ and $ax \leq 1$, and using this comparison in the saturated branches, gives

$$\min\{1, a^2\} \min\{1, x\} \leq a \min\{1, ax\}.$$

Similarly $b \leq \max\{1, b^2\}$: if $b \leq 1$, then $b \leq 1 \leq \max\{1, b^2\}$, while if $b > 1$, then $b \leq b^2 \leq \max\{1, b^2\}$. Splitting on $x \leq 1$ and $bx \leq 1$ gives

$$b \min\{1, bx\} \leq \max\{1, b^2\} \min\{1, x\}.$$

In the branch $x > 1$ and $bx \leq 1$, the needed bound is obtained by first deriving $b < 1$: otherwise $b \geq 1$ would imply $x \leq bx$, contradicting $x > 1$ and $bx \leq 1$.

3. The lower bound follows by chaining the previous inequalities with the block-energy lower bound supplied by [theorem 3](#) and the identification of the coefficient-mass minimax risk in [definition 8](#):

$$\begin{aligned} c_{\beta, p} B^2 \min\{1, x\} &= B^2 \min\{1, a^2\} \min\{1, x\} \\ &\leq B^2 a \min\{1, ax\} \\ &\leq B^2 a \min\{1, y\} \\ &\leq R_n^*(d, \beta, p, B). \end{aligned}$$

4. The projected-SNIPE upper bound at the block-energy scale from [theorem 3](#), together with the upper rescaling above, gives

$$\begin{aligned} \sup_{(G, c) \in \mathcal{M}_{n, d, \beta}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{up}} - \tau_n)^2 \right] &\leq b B^2 \min\{1, y\} \\ &\leq b B^2 \min\{1, bx\} \\ &\leq C_{\beta, p} B^2 \min\{1, x\}. \end{aligned}$$

5. The estimator $\hat{\tau}_n^{\text{up}}$ is an admissible competitor for the minimax problem. The finite-population estimator is measurable, and [assumption 4](#) gives, for every model in $\mathcal{M}_{n,d,\beta}(B)$,

$$|Y_i(z)| \leq \sum_{S \subseteq N_i} |c_{i,S}| \leq B.$$

Therefore

$$|\tau_n| \leq \frac{1}{n} \sum_{i \in V_n} (|Y_i(\mathbf{1})| + |Y_i(\mathbf{0})|) \leq 2B.$$

The projection defining $\hat{\tau}_n^{\text{up}}$ places the estimator in $[-B, B]$, so its pointwise squared loss is bounded by $9B^2$, and hence its risk is uniformly bounded over the class. Since $n \geq 1$, the empty graph with the zero coefficient schedule supplies a member of $\mathcal{M}_{n,d,\beta}(B)$. The definition of $R_n^*(d, \beta, p, B)$ as the infimum over admissible estimators gives

$$R_n^*(d, \beta, p, B) \leq \sup_{(G,c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{up}} - \tau_n)^2 \right].$$

6. The unprojected SNIPE bound is obtained from the block-energy bound in [theorem 3](#):

$$\begin{aligned} \sup_{(G,c) \in \mathcal{M}_{n,d,\beta}(B)} \mathbb{E} \left[(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2 \right] &\leq B^2 \frac{dA_d}{n} \\ &\leq B^2 b x \\ &\leq C_{\beta,p} B^2 x. \end{aligned}$$

7. The score-energy assertion is exactly the local-energy comparison supplied by [theorem 3](#): for every $(G, c) \in \mathcal{M}_{n,d,\beta}(B)$ and every $i \in V_n$,

$$A_i \leq A_d.$$

8. For every $(G, c) \in \mathcal{M}_{n,d,\beta}(B)$, every $i \in V_n$, and every order r with $1 \leq r$, the overlap-count assertion is [lemma 2](#) applied to the graph G , the degree bound d , the order r , and the unit i :

$$\sum_{\ell \in V_n} \binom{|N_i \cap N_\ell|}{r} = \sum_{\substack{S \subseteq N_i \\ |S|=r}} \#\{\ell \in V_n : S \subseteq N_\ell\} \leq d \binom{|N_i|}{r} \leq d \binom{d}{r}.$$

9. Assume $B > 0$ and $d \geq 1$. The block-mixture component of [theorem 3](#), with $m = \lfloor n/d \rfloor$, ρ the active population fraction, and δ the least-favourable perturbation amplitude, gives the two complete-block models M_+ and M_- for every $U = (U_1, \dots, U_m)$ with $|U_b| \leq B/2$. Their graph is the complete-block graph, their coefficient schedules are the corresponding $+1$ and -1 block schedules, and

$$\tau_n(M_+) = \rho\delta, \quad \tau_n(M_-) = -\rho\delta.$$

The same component gives

$$H^2(\Pi_+, \Pi_-) \leq \frac{4\pi^2 m \delta^2}{B^2 A_d}, \quad \frac{1}{2} \leq \rho \leq 1, \quad \frac{A_d}{m} = \frac{dA_d/n}{\rho}.$$

This proves the block-construction assertion and completes the proof. □

Proof of theorem 3. 1. By lemma 1, choose constants c_1, c_2, H depending only on (β, p) , with $0 < c_1 \leq c_2$, and with

$$c_1 \binom{d}{k_\star(d, \beta, p)} \leq A_d \leq c_2 \binom{d}{k_\star(d, \beta, p)} \quad (d \geq 1).$$

Let $H_{\beta, p}$ denote the corresponding uniform raw-representer mass supremum and set

$$\kappa = \min\{(2H_{\beta, p})^{-1}, (4\pi)^{-1}\}, \quad c_{\beta, p} = \min\left\{c_1, \frac{\kappa^2}{16}\right\}, \quad C_{\beta, p} = \max\{16, c_2\}.$$

The representer-mass positivity and $\pi > 0$ give $\kappa > 0$, hence $c_{\beta, p} > 0$, and

$$c_{\beta, p} \leq c_1 \leq c_2 \leq C_{\beta, p}.$$

2. Fix n, D, d, B satisfying the hypotheses. The degree-index condition gives $d \leq n$. The coefficient-mass envelope bounds every potential outcome by B on the same carrier, so $\mathcal{M}_{n, d, \beta}(B) \subseteq \mathcal{M}_{n, d, \beta}^\infty(B)$ carrier by carrier. If $B > 0$ and $d \geq 1$, the one-loop schedule with coefficients B and $-2B$ has uniformly bounded outcomes and coefficient mass $3B > B$, giving strict containment. The equality $R_{n, \ell_1}^\star(d, \beta, p, B) = R_n^\star(d, \beta, p, B)$ is the definition of the coefficient-class minimax risk used here, and the carrier inclusion gives

$$R_{n, \ell_1}^\star(d, \beta, p, B) \leq R_{n, \infty}^\star(d, \beta, p, B).$$

3. The scaled block-prior lower bound gives

$$c_{\beta, p} B^2 \min\left\{1, \frac{dA_d}{n}\right\} \leq R_{n, \ell_1}^\star(d, \beta, p, B).$$

For $d \geq 1$ and $B > 0$, this is obtained with $m = \lfloor n/d \rfloor$, $\rho = md/n$, $t = \min\{1, \sqrt{A_d/m}\}$, and $\delta = B\kappa t$. The choice $\kappa \leq (4\pi)^{-1}$ makes the squared Hellinger distance of the two block priors at most $1/4$. Le Cam's block-prior lower bound then gives $(\rho\delta)^2/4$, and the identities

$$\frac{dA_d}{n} = \rho \frac{A_d}{m}, \quad \frac{1}{2} \leq \rho \leq 1, \quad t^2 = \min\left\{1, \frac{A_d}{m}\right\}$$

convert this to the displayed frontier lower bound. For $B = 0$, and also for $d = 0$, the left side is 0, and the minimax risk is nonnegative.

4. The unclipped SNIPE variance calculation, with the local dependence aggregation controlled by lemma 2, yields on both model classes

$$\sup \mathbb{E}_Z [(\hat{\tau}_n^{\text{SNIPE}} - \tau_n)^2] \leq B^2 \frac{dA_d}{n}.$$

For each model in the coefficient-mass class, and for each model in the uniformly bounded-outcome class, the product Bernoulli design assumption identifies D with the common- p Bernoulli law, and the SNIPE moment identity gives unbiasedness for τ_n .

5. Projection to $[-B, B]$ on the coefficient-mass class and to $[-2B, 2B]$ on the uniformly bounded-outcome class can only improve the modelwise squared error relative to the corresponding target envelopes, while the trivial saturated bounds are $4B^2$ and $16B^2$. Combining these with the unclipped risk bounds gives

$$\sup_{\mathcal{M}_{n,d,\beta}(B)} \mathbb{E}_Z[(\hat{\tau}_n^{\text{up}} - \tau_n)^2] \leq 4B^2 \min\left\{1, \frac{dA_d}{n}\right\},$$

and

$$\sup_{\mathcal{M}_{n,d,\beta}^\infty(B)} \mathbb{E}_Z[(\hat{\tau}_{n,\infty}^{\text{up}} - \tau_n)^2] \leq 16B^2 \min\left\{1, \frac{dA_d}{n}\right\}.$$

Since $4 \leq 16 \leq C_{\beta,p}$, the coefficient-class clipped bound and the bounded-outcome clipped bound have the theorem's constant $C_{\beta,p}$. The minimax risk over the bounded-outcome class is bounded above by the risk of the bounded-outcome clipped SNIPE estimator, giving the displayed minimax upper chain.

6. For every unit i , the exact local score energy is bounded by the complete-block energy:

$$A_i \leq A_d.$$

The same estimate applies on both model classes because it depends only on the degree bound of the carrier graph. Multiplying the energy scale from [lemma 1](#) by d and using $c_{\beta,p} \leq c_1$ and $c_2 \leq C_{\beta,p}$ gives, for $d \geq 1$,

$$c_{\beta,p}d \binom{d}{k_\star(d, \beta, p)} \leq dA_d \leq C_{\beta,p}d \binom{d}{k_\star(d, \beta, p)}.$$

For $d = 0$, both sides reduce by the zero-degree convention.

7. Assume $d \geq 1$ and $d \mid n$, and put $m = n/d$. On the complete directed d -block graph with loops, first consider $B > 0$. The canonical local-linear weights agree with the SNIPE score on each block. The block extremal criterion equals d^2A_d on every block, so summing over m blocks and normalizing yields

$$\frac{B^2A_d}{m}.$$

The fixed-graph coefficient-class risk equals this value. The global coefficient-class risk is bracketed between that fixed-graph risk and the unclipped SNIPE upper bound at the same value; the fixed-graph bounded-outcome risk and the global bounded-outcome risk are obtained by the same comparison, using the carrier-preserving inclusion of the coefficient class in the bounded-outcome class. If $B = 0$, the target B^2A_d/m is zero; the zero schedule gives a nonnegative risk in the fixed-graph coefficient class, and the SNIPE upper bounds and fixed-to-global comparisons force the fixed-graph and global risks in both model classes to be zero. Thus, for all $B \geq 0$, the fixed-graph coefficient-class risk, the fixed-graph bounded-outcome risk, and the corresponding global worst risks all equal B^2A_d/m . Since $n = md$,

$$\frac{B^2A_d}{m} = B^2 \frac{dA_d}{n}.$$

8. Finally suppose $B > 0$ and $d \geq 1$, set $m = \lfloor n/d \rfloor$, $\rho = md/n$, and let δ be the perturbation amplitude. For every $U = (U_1, \dots, U_m)$ with $|U_b| \leq B/2$, the two signed block schedules are members of $\mathcal{M}_{n,d,\beta}(B)$ on the complete-block graph. Their targets are

$$\rho\delta \quad \text{and} \quad -\rho\delta.$$

The block-prior Hellinger calculation gives

$$H^2(\Pi_+, \Pi_-) \leq \frac{4\pi^2 m \delta^2}{B^2 A_d}.$$

The active-share arithmetic gives

$$\frac{1}{2} \leq \rho \leq 1, \quad \frac{A_d}{m} = \frac{dA_d/n}{\rho}.$$

These are exactly the least-favourable family conclusions. □

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